

Module 1 Notes — Business Analysis Foundations

1. SDLC (Software Development Life Cycle)

Definition: The structured stages a system goes through, from idea to retirement.

Standard phases: Planning → Analysis → Design → Development → Testing → Deployment → Maintenance

Where the BA sits: Heaviest involvement in Planning and Analysis (defining the problem, gathering requirements), but stays involved through Design (clarifying intent) and Testing (validating the build matches requirements).

Interview one-liner: "SDLC is the framework for building a system; the BA lifecycle is what I personally do inside that framework to make sure the right thing gets built."

2. SDLC Models

Model	Structure	Best for	Risk
Waterfall	Strictly sequential, one phase finishes before next starts	Fixed-scope, regulated projects (banking, government, healthcare compliance)	Expensive to change requirements once dev has started
Agile	Short iterative cycles (sprints), requirements evolve	Most modern SaaS/product companies	Needs disciplined backlog management or scope creep happens
Iterative	Repeated cycles, each adds more functionality to a working system	Complex systems built incrementally	Early iterations may not reflect final architecture
V-Model	Waterfall + a mirrored testing phase for every dev phase	Safety-critical systems (medical devices, aviation)	Rigid, same downsides as Waterfall
Spiral	Waterfall + Iterative + explicit risk analysis every loop	Large, high-risk, high-cost projects	Slower, more overhead, needs risk-analysis expertise

Common interview trap: "Which is best?" → There's no universal answer — it depends on how well-defined requirements are upfront, regulatory needs, and project size/risk. Never answer with just one model's name.

3. The BA Lifecycle

Enterprise Analysis — understand the *business problem* before jumping to a solution

Requirements Planning — decide who to talk to, which technique, and timeline

Elicitation — gather requirements (interviews, workshops, surveys, observation, document review)

Analysis & Documentation — turn raw input into BRD/FRD/user stories

Solution Assessment & Validation — confirm the delivered solution actually solves the original problem
Beginner mistake to avoid: jumping straight to step 4 (writing documents) without real elicitation (steps 1–3)
— this produces documents built on assumptions instead of evidence.

4. Requirement Gathering Techniques

Technique	Strength	Weakness
Interviews (1:1)	Deep, individual insight; good for sensitive topics	Time-consuming to scale across many stakeholders
Workshops / JAD sessions	Resolves conflicting needs fast, multiple views at once	Can be dominated by louder voices in the room
Surveys/Questionnaires	Scales to many people cheaply	Low depth, no follow-up possible
Observation ("job shadowing")	Reveals things people forget to mention or don't realize they do	Time-intensive, people may behave differently when watched
Document Analysis	Fast way to understand existing process/systems	Documents may be outdated or not reflect real practice
Prototyping	Surfaces requirements people can't articulate cold, gets concrete reactions	Requires a draft/mockup to already exist

My own experience (Week 1): used 1:1 interviews across 3 stakeholder types (manager, frontline staff, compliance) and found that a single interview gives an incomplete picture — the receptionist interview surfaced a workaround the manager didn't know about, and the compliance interview surfaced a risk nobody had mentioned. **Triangulating across roles is what surfaces the real picture.**

5. Stakeholder Analysis

Two core tools:

RACI Matrix — for each task/decision, define who is Responsible, Accountable, Consulted, Informed. Only one person/role should ever be *Accountable* per task — that's the single point of sign-off.

Power/Interest Grid — plots stakeholders on two axes:

Manage Closely (high power, high interest) — most engagement, sign-off matters most

Keep Satisfied (high power, low interest) — update periodically, don't overwhelm with detail

Keep Informed (low power, high interest) — most affected day-to-day, needs regular updates

Monitor (low power, low interest) — minimal effort, light-touch communication

Applied example (Healthcare project): Ops Manager and Chief Medical Officer = Manage Closely; IT Director and Compliance Officer = Keep Satisfied; front-desk staff, billing, nurses = Keep Informed; patients and vendor = Monitor.

6. BRD vs FRD vs SRS

Document	Answers	Audience	Example
BRD (Business Requirements Document)	<i>Why</i> are we doing this?	Business stakeholders/sponsors	"Reduce missed appointments by enabling online booking."
FRD (Functional Requirements Document)	<i>What</i> should the system do?	Delivery/dev team	"System shall let a patient select an open slot and get an SMS confirmation within 60 seconds."
SRS (Software Requirements Specification)	<i>How</i> should it behave, technically?	Developers/QA	Includes functional + non-functional requirements: performance, security, uptime targets.

Memory hook: BRD (business problem) → FRD (system behavior) → SRS (technical detail). In practice, many companies merge FRD and SRS into one document — know the theoretical distinction, but don't be thrown if a real job blends them.

7. Gap Analysis

Definition: Compare As-Is (current state) vs To-Be (desired state); the difference is the gap the project must close.

4-column structure to always use:

As-Is

To-Be

The Gap (specific deficiency)

Priority/Impact

Applied example (Healthcare project): Found 6 gaps — from a technical dedup issue to a compliance/regulatory unknown — and prioritized patient-facing failures and legal/compliance risk above pure efficiency gains, even though efficiency gains are easier to explain to non-technical stakeholders.

8. SWOT Analysis

Strengths — internal advantages

Weaknesses — internal shortcomings

Opportunities — external factors that could help

Threats — external factors that could hurt

Key distinction from Gap Analysis: SWOT is broader and includes external factors (market, competitors, regulation); Gap Analysis is narrowly about current-state vs desired-state for a specific process. Used earlier in a project, often at the business-case stage, before Gap Analysis gets specific.

9. Root Cause Analysis (RCA)

Two main techniques:

5 Whys — ask "why" repeatedly (usually 5x) until you reach the actual root cause, not just a symptom.

Example: *Why do duplicate profiles happen?* → Search misses patients → *Why?* → Exact-match only → *Why?* → No fuzzy matching built in → *Why?* → Never prioritized in original system design → *Why?* → No one owned data-quality requirements at build time. (**Root cause: no ownership of data-quality requirements, not just "bad search."**)

Fishbone (Ishikawa) Diagram — categorizes possible causes into buckets: People, Process, Technology, Environment — useful when a problem could stem from multiple directions at once.

Why this matters: fixing a symptom (e.g., "add a warning message") without finding the root cause (e.g., "no fuzzy match logic") means the problem resurfaces in a different form later.

10. Business Process Mapping

Definition: Visual documentation of how a process actually works, step by step — usually with **swimlanes** showing which role performs each step.

Why it matters to a BA: a flowchart reveals things a narrative description hides — bottlenecks, handoff points where errors creep in, and steps that don't need to exist at all.

Tools: draw.io (free) and Visio (enterprise standard) — covered hands-on in Module 7.

Applied example: mapped the as-is booking flow — patient calls → phone search (exact-match only) → branch on match found/not found → slot booked → manual reminder — and the *branch point* (what happens when no match is found) is exactly where the duplicate-profile risk lives. Seeing it as a flow, not a paragraph, made the risk point obvious.

Self-Test (close this file and answer without looking)

What's the difference between the BA lifecycle and the SDLC?

Name the 5 SDLC models and one real-world scenario where each fits best.

What's the difference between "Keep Satisfied" and "Keep Informed" on a power/interest grid?

In one sentence each, distinguish BRD, FRD, and SRS.

What are the 4 columns of a Gap Analysis, in order?

What's the difference between SWOT and Gap Analysis?

Walk through a 5 Whys example for a problem of your choosing.

Why does a process map sometimes reveal things a written description doesn't?

(Answers are all in the sections above — if you can't answer one from memory, that section needs another pass before moving to Week 2.)